Source: "Bridge Crane Path Planning System Based on LabVIEW" — IEEE conference paper, 2019. https://ieeexplore.ieee.org/document/8941516

Core: The paper mainly goes over pathfinding/movement as its core

What it does: The paper does pathfinding for a bridge crane and does object avoidance in both two or three dimensions. They mainly use labVIEW to compute this path for the crane before moving.

How it handles the core feature: For the 2D mode the system does a grid and runs A* pathfinding algorithm to build an obstacle free route. If the system cannot do A* pathfinding well enough to connect the start and end point the system will start to pathfinding on a 3D level. This will use a laser to detect obstacles to maintain clearance with the laser being on the payload itself to help more accurately avoid the obstacles in its path.

What I learned from it: The main thing I learned from this paper is that the A* pathfinding solution is the correct choice to move into and focus on over other pathfinding methods. Also puts into perspective what should be done if A* can't pathfind between two points. Would have to talk to the team about what to do and if we should do a Z-axis implementation and then running A* or just have a dedicated 3D pathfinding method.